

PS SATCHITH

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Research Summary

Integrated MSc Physics student working on the computation of quantum many-body systems. My research so far spans two areas: quantum information at the level of photons, where I simulated quantum algorithms using linear optics during a national research fellowship, and first-principles simulation, where I run ab initio molecular dynamics in CP2K. I am now learning and exploring the directions I want to build on next: quantum simulation, quantum-computing methods and machine learning for physical systems.

Research Interests

- Quantum simulation and quantum-computing methods for physical systems
- Quantum information and photonic quantum computing (quantum algorithms, quantum walks, LOQC)
- Machine learning for physical systems and atomistic simulation
- Ab initio / first-principles molecular dynamics

Education

Amrita Vishwa Vidyapeetham, Coimbatore, Integrated MSc Physics 2022 - 2027

- CGPA: 8.11/10
- Relevant Coursework: Quantum Mechanics I & II, Condensed Matter Physics, Atomic & Molecular Spectroscopy, Mathematical Physics I & II, Electrodynamics, Wave Optics, Nuclear & Particle Physics, General Relativity, Special Relativity, Physics of Compact Stars.

Palghat Lions School, Higher Secondary (CBSE) 2019 - 2021

- Percentage: 86.4/100

Research Experience

IAS-INSANA-Joint Academies Summer Research Fellow Apr - Jul 2025

Under Dr. Atanu Bhattacharya

- Project: *Foundations and Applications of Photonic Qubits - Implementing Quantum Algorithms using Linear Optical Quantum Computing.*
- Studied the classical-to-quantum optics transition; simulated the Deutsch algorithm using polarization/path encoding.
- Designed an optical circuit for Gaussian 2-qubit state initialization using SPDC and LOQC components; prepared a detailed technical report.
- Awarded a competitive national research fellowship (jointly by the three Indian science academies).

REYES 2025 Mentorship Programme, UC Berkeley Jul - Sep 2025

- Authored a pedagogical review, *Foundations of Photonic Quantum Computing*, submitted to UC Berkeley as a programme outcome.
- Mentored two students through the foundations of photonic quantum computing.

Conference Presentations

- **Poster:** "Ab-Initio Molecular Dynamics Study of Sodium Perchlorate in Water." *GITAM Chemistry Conference*, February 2026.

Selected Projects

Ab Initio Molecular Dynamics Study of Sodium Perchlorate (NaClO_4) in Water

2026

- Performed AIMD simulations (CP2K) of aqueous NaClO_4 to investigate Na^+ solvation structure, water hydrogen bonding, and short-time dynamics.
- Identified weaker ion-solvent interactions relative to Li^+ (enhanced mobility); quantified diffusion from trajectories and linked solvation environment to electrolyte performance for sodium-ion batteries.
- **Tools:** CP2K, RDF & diffusion analysis (Python), $\LaTeX$.

Working Model of LIGO

2025

- Built a Michelson interferometer with Fabry-Pérot cavities; acquired phototransistor signals via Arduino and recovered “strain vs. time” using bandpass filtering and FFTs in Python.
- Won the Physics Category Prize at SCIVIO 2025 (university science fest).
- **Skills:** signal processing, data acquisition, programming, experimental optics.

Technical Skills

- **Simulation & Computing:** CP2K (DFT-based AIMD), Python-based trajectory analysis, PyTorch (basics).
- **Programming:** Python (NumPy, SciPy, Matplotlib, Pandas), Bash, Git/GitHub.
- **HPC & environment:** Linux, SLURM, GPU computing.
- **Analysis & other:** RDF / coordination / diffusion analysis, OriginLab, $\LaTeX$, Qiskit (basic).

Workshops & Schools

International Workshop on Theoretical Physics (IWTP-2024), Amrita

April 2024

- Lectures by Prof. Subhendra Mohanty (IIT-K) and Prof. Sergei B. Popov (ICTP, Trieste) on cosmology, attosecond solitons, and quantum computing.

Pravega 2-Day Astrophysics Workshop, IISc Bangalore

November 2023

Certifications

- Introduction to Quantum Computing: Quantum Algorithms and Qiskit - NPTEL (2024).
- Basics of Quantum Information - IBM (2025).
- Introductory Course on Quantum Computing - I-HUB QTF, IISER Pune (2025).

Extra-Curricular Activities

- Represented the university and placed 5th in the 38th AIU South-Zone Inter-University Group Music competition.
- Volunteer, Anokha 2024 - 12th National-Level Techfest, Amrita Vishwa Vidyapeetham.
- Qualified for the Sastra Pratibha Matsaram State-Level Science Camp (2015), Kerala Forest Research Institute.